Six-hour PM2.5 forecasting in Beijing with gradient boosting and leak-free backtesting

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***Abstract*: We forecast hourly PM2.5 at the US Embassy in Beijing one, six and 24 hours ahead, with six hours as the headline horizon, using only information available when the forecast is made. A model sees 52 inputs in six groups: PM2.5 history, rolling statistics, the calendar of the target hour, weather from the hour before the forecast, derived weather, and physics features for stagnation, southerly transport and humid build-up taken from studies of Beijing winter pollution episodes. XGBoost is tuned by random search, averaged over five seeds and blended with ridge regression. Every choice is made on three expanding-window validation years (2011 to 2013) with a 24-hour embargo, and the 2014 test year is scored once. At six hours the blend reaches an RMSE of 51.96 $\mu g\,m^{-3}$, MAE 33.93 and R^2 0.690 on 2014, against 61.22, 38.67 and 0.569 for persistence: 15.1% lower RMSE, close to the 14.4% measured during validation. After PM2.5 history, weather adds the most accuracy, and a raw target beats log1p, which biases forecasts low by 12.6 $\mu g\,m^{-3}$. Errors concentrate in pollution episodes: hours above 150 $\mu g\,m^{-3}$ are 21.2% of validation hours but carry 50.6% of the squared error. A mixture of experts with a learned episode gate lowers validation MAE by 1.2% in every seed set, but its RMSE gain is within seed noise; with a perfect gate the same experts would cut RMSE by 30%, which locates the remaining limit in predicting when episodes start and end.**

I. Introduction

Beijing's PM2.5 alternates between clean air and severe multi-day episodes driven by emissions and weather [2]. Forecasting it a few hours ahead supports health warnings, but a forecast is only useful if it beats the simplest rule, that PM2.5 stays at its current value (persistence), and if its accuracy is measured on data the model has never influenced [12], [14].

This project compares gradient boosting (Phase 1, this report) with recurrent and convolutional networks (Phase 2) on the UCI Beijing PM2.5 dataset [1]. A companion report [3] covers the exploratory data analysis (EDA) and preprocessing; Section III summarises the findings that shape the models. We chose six hours as the headline horizon before training any model: one hour ahead, copying the current value is already wrong by only 13.1 $\mu g\,m^{-3}$ on average, while 24 hours ahead it is wrong by 69.8 and the inputs carry little signal [3].

Our contributions are:

- a direct forecasting setup with explicit look-ahead rules, an embargo between training and evaluation, and a test year guarded so that it can be scored only once (Section IV);
- physics-informed features from the Beijing episode literature and a five-seed ablation of every feature group and target form (Sections V and VII-C);
- a tuned XGBoost and ridge blend that beats persistence by 15.1% RMSE at six hours on the unseen year 2014 (Section VII);
- an error analysis by season and pollution episode, and a mixture-of-experts extension that shows where the remaining error comes from (Sections VII-E and VIII).

II. Related work

A. PM2.5 meteorology in Beijing

Liang et al. [2] introduced the dataset used here and showed that weather explains much of the variation in Beijing's PM2.5. Zhong et al. [5] split winter episodes into a transport stage, when pollution arrives on southerly winds, and a cumulative stage under weak southerly or calm winds, a shallow boundary layer and relative humidity above 80%, and attribute over 70% of the cumulative explosive growth to meteorological feedback. Huang et al. [6] define a stagnant day by 10 m wind below 3.2 m/s, 500 hPa wind below 13 m/s and no precipitation. These two studies are the source of our physics features.

B. Forecasting models

Persistence and seasonal-naive forecasts are the standard reference points for short-horizon forecasting [7]. Gradient-boosted trees [8] handle nonlinear interactions, missing inputs and mixed feature scales without preprocessing, which makes them a strong tabular baseline. Recurrent networks such as the LSTM [9] read raw input windows instead of hand-built lags and are the Phase 2 candidates. Wang and Du [10] compare persistence, seasonal-naive, ridge, elastic net and five-seed XGBoost on 12 Beijing stations (2013 to 2017) with PM10 as an extra input. Their best R^2 is 0.953, 0.660 and 0.242 at 1, 6 and 24 hours, with XGBoost best at six hours; we use their results as an external reference for the shape of the problem.

C. Multi-step strategy

A forecast h steps ahead can come from one model applied recursively, from one model per horizon (direct), or from one model with several outputs [11]. The recursive strategy feeds forecasts back as inputs and accumulates their errors; we use the direct strategy, one model per horizon.

D. Evaluation and leakage

Time series must be evaluated on data after the training period [12], [13]. Leakage, any route by which information from the evaluation period reaches training, inflates scores

[14]. Overlapping labels are removed with a gap between training and evaluation, which López de Prado calls purging [15].

E. Combination and experts

Combining forecasts with weights chosen to minimise squared error often beats the individual forecasts [16]. A mixture of experts [17] goes further and lets a gating model weight each expert per input. When a model is trained on log1p of the target, the naive back-transform underestimates the mean; Duan's smearing factor [18] corrects it.

III. Data and exploratory findings

The dataset [1] holds hourly PM2.5 measured at the US Embassy together with weather from Beijing Capital International Airport, 2010-01-01 to 2014-12-31 (43 824 hours). Only PM2.5 has missing values (2 067 hours, 4.72%); the cumulative wind column is decoded into hourly speed, and the missing target is never filled [3]. The years 2010 to 2013 are used for development; 2014 is the test year. Table I lists the EDA findings, all measured on 2010 to 2013, that shaped the models.

TABLE I
EDA findings on 2010 to 2013 and the design choices they led to [3]

Finding	Design choice
Persistence MAE 13.1, 40.8, 69.8 at 1, 6, 24 h; autocorrelation of log1p(pm25) 0.96, 0.74, 0.33	Six-hour headline; lags 1 to 48 h; rolling statistics
Opposite daily cycles: winter median peaks at 21:00, summer at 08:00	Hour and month of the target hour as sine and cosine
North-west wind cleanest (median 31), calm and south-east dirtiest (100, 93)	Wind direction, speed and NW wind run
First three hours of a NW spell: median change −12.2% in six hours	Hours since the wind direction changed
Pressure rise above 2 hPa in 3 h: median change −21.4%	Pressure tendency over 3 and 24 h
Relative humidity is the strongest weather signal (Spearman 0.54)	Relative humidity, humid hours
322 episodes above 150, slow build (median 55 h), fast clearing (14 h)	Physics features; episode-hour scores
Lunar New Year peaks 7 to 14 times the usual level	Lunar New Year flag
pm25 skew 1.81	Raw and log1p targets compared

IV. Problem formulation and evaluation protocol

A. Direct forecasting

Let y_t be PM2.5 at hour t, the forecast origin. For each horizon $h \in \{1, 6, 24\}$ a separate model predicts

$$\hat{y}_{t+h} = f_h(\mathbf{x}_t), \tag{1}$$

where $\mathbf{x}_t$ contains PM2.5 up to hour t, weather up to hour $t-1$, and the calendar of the target hour $t+h$, which is known in advance. Weather is taken one hour before the origin because another UCI release of the same station [4] has identical weather but PM2.5 shifted by one hour, and we cannot tell which clock is right; with a one-hour lag no weather input can share an hour with the target either way.

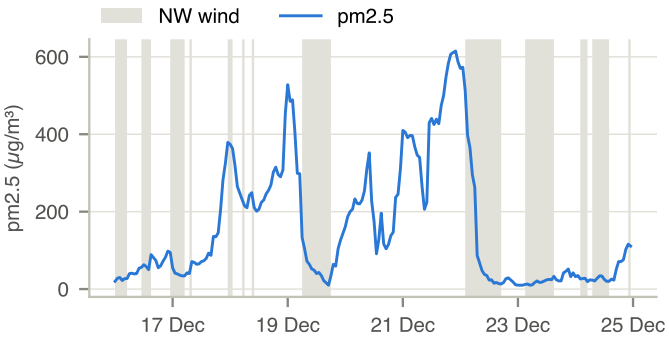


Fig. 1. Two episodes in December 2010: slow build-up, then collapse when north-west wind (shaded) arrives [3].

B. Folds and embargo

Fold k evaluates on calendar year Y_k and trains on all hours before 1 January of Y_k minus a 24-hour embargo, so no training target falls in the evaluation year for any $h \le 24$ [15]. The validation folds cv1, cv2 and cv3 evaluate 2011, 2012 and 2013 with one, two and three training years; the test fold trains on 2010 to 2013 and evaluates 2014. An evaluation hour is kept only if the origin and the target both lie inside the evaluation year, the target was observed, and both baselines can be computed. Every model is therefore scored on the same hours: 24 417 pooled validation hours at six hours.

C. Metrics and selection

With errors $e_i = \hat{y}_i - y_i$ over n hours,

$$\mathrm{MAE} = \frac{1}{n}\sum_i |e_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i e_i^2}, \tag{2}$$

$$R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}, \qquad \mathrm{skill} = 1 - \frac{\mathrm{RMSE}}{\mathrm{RMSE}_{\mathrm{persistence}}}, \tag{3}$$

all in $\mu\mathrm{g\,m^{-3}}$ on the original scale, plus the bias $\bar{e}$. Every choice (hyperparameters, target form, ridge penalty, blend weights) minimises RMSE pooled over cv1 to cv3. Both models train on squared error, which RMSE measures, and on a fixed set of hours RMSE and R^2 rank models identically. Because choices are made and scored on the same folds, validation scores are slightly optimistic; the test year is the honest estimate.

D. Score-once test year

All settings were frozen before 2014 was scored, once, on 2026-10-19. The scoring step refuses to run if test scores already exist and stores a SHA-256 fingerprint of the configuration and the chosen settings, so a second look at the test year cannot happen silently.

V. Features

Table II lists the 52 inputs at six hours. Two parts depend on the fold or horizon and are added per fold: the calendar of the target hour, and `pm_anom`, PM2.5 now minus the mean

TABLE II
FEATURE GROUPS (52 COLUMNS AT $h = 6$)

Group	n	Columns
Lags	10	PM2.5 now and 1, 2, 3, 6, 12, 24, 48 h ago; missing flag now; missing hours in the last 24 h
Rolling	15	mean, std, max over 3, 6, 12, 24 h; change over 1 and 3 h; `pm_anom`
Calendar	7	target hour and month as sin and cos, day of week, weekend, Lunar New Year eve and day
Weather	11	dew point, temperature, pressure, wind speed, cumulative wind, rain, snow, four wind directions; hour $t-1$
Derived	5	relative humidity, pressure change over 3 and 24 h, hours since the wind direction changed, NW wind run over 24 h
Physics	4	stagnant hours in 24 h, current stagnant run, SE wind run over 24 h, humid hours in 24 h

PM2.5 for that hour of day in the fold's training years. Relative humidity uses the Magnus formula [19].

The physics group turns the episode literature into hourly inputs. An hour is stagnant when the wind is weak and dry,

$$s_t = \mathbf{1}[\, ws_t < 3.2 \text{ m/s} \wedge \text{rain}_t = 0\,], \tag{4}$$

the surface part of the stagnation definition in [6], applied per hour. From it, $\texttt{stag24} = \sum_{k=1}^{24} s_{t-k}$ counts stagnant hours and `stag_run` is the length of the current stagnant run. `se_ws24` sums south-east wind speed over 24 hours (the transport stage of [5]) and `humid24` counts hours with relative humidity above 80% (the cumulative stage). Trees split on single values and cannot build counts or run lengths themselves, so these columns add memory the weather group lacks.

Every feature at origin t uses rows up to t only. A test deletes every row after an origin and checks that all feature rows up to it are unchanged; further tests check that weather inputs lag the origin, that training targets respect the embargo, that adding later years never changes a fold, and that the hour-of-day climatology ignores evaluation labels.

VI. MODELS

A. Baselines

Persistence forecasts $\hat{y}_{t+h} = y_t$ and seasonal-naive forecasts $\hat{y}_{t+h} = y_{t+h-24}$, the same hour one day earlier. Both use the causal input copy of PM2.5, carried forward for at most three hours across gaps [3].

B. Ridge regression

Ridge regression [20] minimises $\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \alpha\|\boldsymbol{\beta}\|^2$. The penalty stabilises coefficients when inputs are collinear, as dew point, temperature and pressure are ($|r|$ 0.78 to 0.83 [3]). PM2.5 level columns are log-transformed, missing values are filled with training medians and all columns are standardised, all fitted on training rows. The penalty $\alpha \in \{0.1, 1, 10, 100, 1000\}$ and the target (raw or log1p) are chosen together; the raw target with $\alpha = 10$ wins at one and six hours, $\alpha = 1000$ at 24 hours.

C. XGBoost

XGBoost [8] builds an additive model of M regression trees, $\hat{y} = \sum_{m=1}^{M} f_m(\mathbf{x})$, and adds trees one at a time to minimise

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_m \Omega(f_m), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2, \tag{5}$$

where T is the number of leaves and $\mathbf{w}$ the leaf values. With a second-order expansion of the loss, g_i and h_i its first and second derivatives at the current prediction, and G_j, H_j their sums over the rows in leaf j, the best leaf value and the gain of a split into left and right children are

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{6}$$

$$\text{gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma. \tag{7}$$

For XGBoost's squared-error objective $\frac{1}{2}(\hat{y} - y)^2$, $h_i = 1$, so H_j counts rows and `min_child_weight` sets the minimum leaf size.

Training uses learning rate 0.05 and up to 2 000 trees. Early stopping (100 rounds) watches the last 90 days of the training span, with an h-hour gap before them so no fitting target falls inside, and the model is then refitted on the whole span with the chosen number of trees. A random search draws 20 candidates from depth $\{3, 4, 5, 6, 8\}$, `min_child_weight` $\{1, 5, 10, 20\}$, row subsample $\{0.6, 0.8, 1.0\}$, column subsample $\{0.5, 0.7, 0.9\}$ and $\lambda \in \{0.1, 1, 10\}$. At six hours the search chose depth 4, `min_child_weight` 20, row subsample 1.0, column subsample 0.5 and $\lambda = 1$, with 127 trees on average over the folds.

D. Target forms

Each candidate is tried with two target forms: the level y_{t+h}, and the change $y_{t+h} - y_t$, added back to the current value. A tree predicts averages of training targets, so a level model cannot forecast beyond the largest value it has seen, while a change model starts from the current value. The change form won at one hour (RMSE 22.54 against 23.81) and 24 hours (83.30 against 83.48), the level form at six hours (56.84 against 57.18). We also compare training on $z = \log(1 + y)$: since $\mathbb{E}[\exp z] \geq \exp \mathbb{E}[z]$, undoing the log underestimates the mean, which Duan's smearing factor $s = \frac{1}{n}\sum_i \exp(z_i - \hat{z}_i)$ [18], computed on the early-stopping period, corrects as $\hat{y} = s\exp(\hat{z}) - 1$.

E. Ensembles

The seed ensemble averages five XGBoost models that differ only in their random seed, which reduces the variance from column subsampling. A spike expert uses the same settings with the quantile loss $\rho_q(u) = u\,(q - \mathbf{1}[u < 0])$ at $q = 0.9$, so it aims high on purpose. The blend

$$\hat{y} = a\,\hat{y}_{\text{xgb}} + b\,\hat{y}_{\text{spike}} + c\,\hat{y}_{\text{ridge}}, \quad a, b, c \geq 0,\ a + b + c = 1, \tag{8}$$

takes its weights from a 0.1 grid by pooled RMSE [16]. It chose $(a, b, c) = (0.7, 0, 0.3)$, $(0.6, 0, 0.4)$ and $(0.3, 0, 0.7)$ at 1, 6 and 24 hours; the spike expert never received weight.

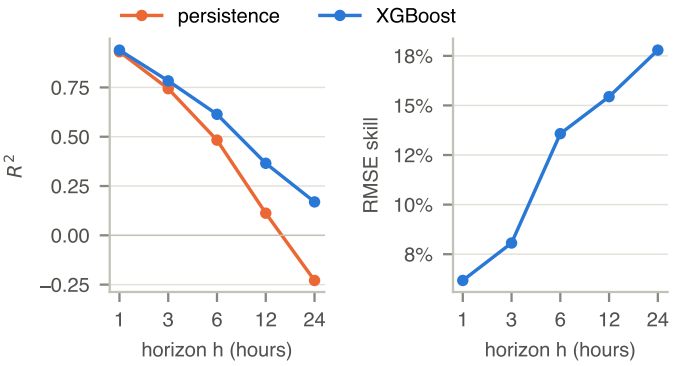


Fig. 2. Validation R^2 of persistence and XGBoost (left) and XGBoost RMSE skill over persistence (right) by horizon. Horizons 3 and 12 reuse the six-hour settings.

VII. Results

Table III shows validation and test scores. On both, the blend has the lowest RMSE at every horizon, and XGBoost has a slightly lower MAE at one hour and, on the test year, at six hours. Seasonal-naive is the weakest forecast except at 24 hours, where it equals persistence.

A. Generalisation to 2014

2014 is an easier year than the validation years: persistence alone explains 56.9% of the six-hour variance against 48.3%. Absolute errors are therefore lower, and the fair comparison is skill. The blend's six-hour RMSE skill is 15.1% on 2014 against 14.4% on validation (MAE skill 12.3% against 9.5%), so the gain carried over to unseen data. Our R^2 of 0.951, 0.690 and 0.241 at 1, 6 and 24 hours is close to the 0.953, 0.660 and 0.242 reported for 12 stations with PM10 [10]; the data differ, so this compares the shape of the problem, not the models.

B. Choice of horizon

Fig. 2 sweeps the horizon with the six-hour settings. Skill keeps rising with h because persistence degrades faster than the model, so skill alone does not pick a horizon. R^2 completes the argument: at one hour persistence already explains 93% of the variance, and beyond six hours XGBoost explains less than half (0.365 at 12 hours). Six hours is the longest horizon at which the model explains most of the variance (0.614) while clearly beating persistence (13.6% RMSE skill). The headline was fixed from the EDA before training; this sweep confirms it.

C. Ablation

Fig. 3 adds feature groups one at a time at six hours; each point is the mean forecast of five seeds, because a single fit moved by 0.74 RMSE from column order alone. Weather brings the largest gain after the PM2.5 history (60.16 to 58.02), derived weather and the physics group add 0.85 and 0.36 more. Table IV compares target forms with all features. Training on log1p biases forecasts low by 12.6 µg m^{-3}, as the inequality above predicts; smearing overcorrects to +7.1, and the raw level is best on RMSE.

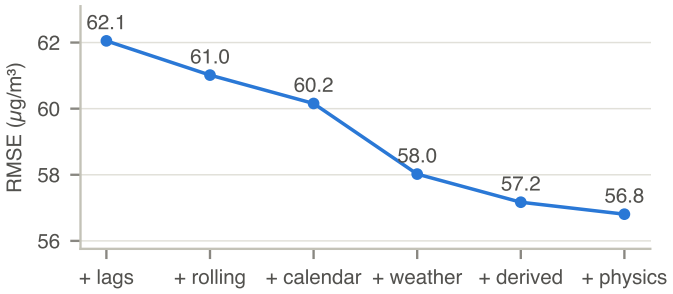


Fig. 3. Validation RMSE at six hours as feature groups are added (XGBoost, mean of 5 seeds).

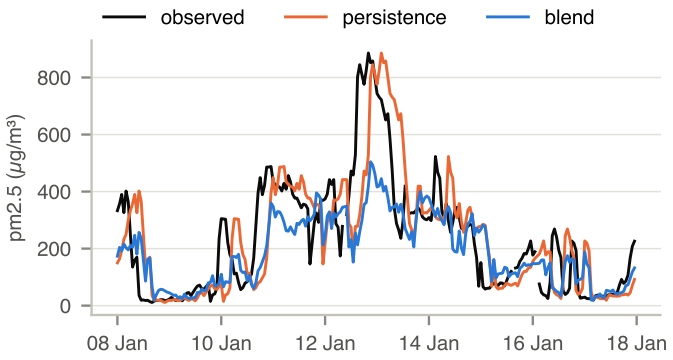


Fig. 4. Six-hour forecasts in January 2013 (validation year of cv3), plotted at their target time.

D. Feature importance

Table V ranks inputs by permutation importance [21]: the rise in validation RMSE when one column is shuffled, fitted on cv3 and measured on 2013. `pm_anom` dominates; it carries the current level and how unusual it is for that hour, so it overlaps with `pm_0`. Two of the four physics features rank third and sixth, ahead of every raw weather column, and the three-hour pressure tendency and south-east wind follow.

E. Where the errors are

Table VI splits the six-hour validation error by season and by episode hours, where PM2.5 at the target hour exceeds 150 µg m^{-3}. Winter error is 1.7 times the spring error. Episode hours are 21.2% of the hours but carry 50.6% of the blend's squared error, and the blend predicts them 55.3 µg m^{-3} too low on average; the top 5% of hours alone carry 53.9% of the squared error. The spike expert is best on episode hours (RMSE 75.97) but far worse overall (82.70), which is why the blend gave it no weight. Fig. 4 shows the typical failure: the blend follows the build-up but lags the sharp rises and clearings.

F. Air quality categories

To read the forecasts as warnings, we map them to the six PM2.5 categories of China's HJ 633-2012 [22] (breakpoints 35, 75, 115, 150 and 250 µg m^{-3}). The standard uses 24-hour means, so applying it to single hours is for illustration only, and the test year is not used because it may be scored only

TABLE III
SCORES BY HORIZON. VALIDATION: POOLED OVER 2011 TO 2013, USED FOR EVERY CHOICE. TEST: 2014, SCORED ONCE. RMSE AND MAE IN $\mu g\,m^{-3}$.

	$h = 1$			$h = 6$			$h = 24$		
Model	RMSE	MAE	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2
Validation, 2011 to 2013									
Persistence	24.02	12.82	0.931	65.77	40.23	0.483	101.33	69.34	−0.229
Seasonal-naive	101.36	69.41	−0.229	101.42	69.42	−0.230	101.33	69.34	−0.229
Ridge	23.07	13.15	0.936	57.66	38.27	0.602	82.21	60.30	0.191
XGBoost (5 seeds)	22.53	**12.21**	0.939	56.81	36.46	0.614	83.63	60.56	0.163
Spike expert ($q = 0.9$)	30.68	20.85	0.887	82.70	62.21	0.182	136.54	109.23	−1.231
Blend	**22.45**	12.23	**0.940**	**56.33**	**36.41**	**0.621**	**81.97**	**59.93**	**0.196**
Test, 2014									
Persistence	21.98	11.92	0.945	61.22	38.67	0.569	99.02	67.49	−0.121
Seasonal-naive	98.87	67.36	−0.121	98.74	67.30	−0.121	99.02	67.49	−0.121
Ridge	21.41	12.01	0.947	54.20	36.11	0.662	81.63	60.13	0.238
XGBoost (5 seeds)	20.74	**11.27**	0.951	52.21	**33.79**	0.687	83.08	60.23	0.211
Spike expert ($q = 0.9$)	28.90	19.43	0.904	76.09	56.28	0.334	131.73	105.25	−0.984
Blend	**20.64**	11.30	**0.951**	**51.96**	33.93	**0.690**	**81.49**	**59.78**	**0.241**

TABLE IV
TARGET FORM AT SIX HOURS (XGBOOST, 5 SEEDS, VALIDATION)

Target	RMSE	MAE	R^2	Bias
Level, log1p	59.96	37.03	0.570	−12.63
Level, log1p + smearing	58.11	37.19	0.596	7.10
Level, raw	**56.81**	36.46	**0.614**	1.71
Change, raw	58.05	**36.42**	0.597	0.89

TABLE V
TOP 10 FEATURES AT SIX HOURS (CV3, EVALUATED ON 2013)

Feature	Group	Permutation ($\mu g\,m^{-3}$)	Gain share
`pm_anom`	rolling	19.32	28.4%
`pm_0`	lags	2.35	9.2%
`stag_run`	physics	1.95	1.5%
`hour_cos`	calendar	1.65	2.0%
`pm_mean24`	rolling	1.49	2.9%
`stag24`	physics	1.39	2.4%
`pres_diff3`	derived	1.11	1.1%
`wd_se`	weather	1.04	1.5%
`month_cos`	calendar	0.73	1.3%
`ws`	weather	0.73	1.1%

TABLE VI
SIX-HOUR VALIDATION RMSE BY SUBSET ($\mu g\,m^{-3}$)

Subset	Hours	Persistence	XGBoost	Blend
Winter (DJF)	6 064	88.79	77.35	75.57
Spring (MAM)	6 000	51.08	44.85	45.08
Summer (JJA)	6 051	51.57	45.47	45.68
Autumn (SON)	6 302	64.33	53.34	53.48
Episode ($y > 150$)	5 169	89.21	89.39	87.08
All	24 417	65.77	56.81	56.33

once. On the validation years (Table VII) the blend lowers the MAE from 41.4% to 37.5% of the mean level (97.1 $\mu g\,m^{-3}$) and is within one category more often, but it hits the exact category slightly less often than persistence, which is exact whenever the level does not change.

TABLE VII
HJ 633-2012 CATEGORIES AT SIX HOURS, VALIDATION YEARS (%)

	Persistence	Blend
MAE as share of mean level	41.4	**37.5**
Exact category	**49.0**	47.9
Within one category	84.1	**87.3**
Correct above-75 call	79.3	**80.4**
Correct above-150 call	86.1	**87.7**

VIII. EXTENSION: MIXTURE OF EXPERTS

The spike expert failed because one fixed weight applies to every hour. A mixture of experts [17] can weight it hour by hour. We built one at six hours on the validation years only; 2014 is not used.

A. Design

The normal expert is the frozen blend. The spike expert is XGBoost with squared error trained only on hours whose target exceeds 150 $\mu g\,m^{-3}$ (1 952, 3 735 and 5 358 training hours in cv1 to cv3). A gate, an XGBoost classifier with the same inputs and settings, estimates $p = P(y_{t+6} > 150 \mid \mathbf{x}_t)$. Two combinations are compared:

$$\hat{y}_{\text{gate}} = \hat{y}_{\text{normal}} + \lambda\, p\, (\hat{y}_{\text{spike}} - \hat{y}_{\text{normal}}), \tag{9}$$

$$\hat{y}_{\text{split}} = (1 - p)\, \hat{y}_{\text{quiet}} + p\, \hat{y}_{\text{spike}}, \tag{10}$$

with λ chosen on a 0.1 grid (0.1) and a quiet expert trained on hours at or below 150. If the experts estimate the conditional means of their groups and p is calibrated, $\hat{y}_{\text{split}}$ is the conditional mean $\mathbb{E}[y_{t+6} \mid \mathbf{x}_t]$ by the law of total expectation, the quantity RMSE rewards.

B. Gate quality

The gate separates episode hours well (Table VIII) and beats the current level alone everywhere, most of all where it matters: onsets (episode hours with PM2.5 at most 150 at the origin) and clearings (origin above 150, target at most 150).

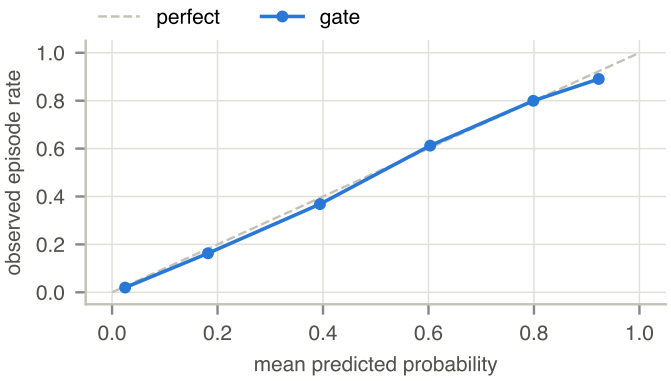


Fig. 5. Gate calibration: mean predicted probability against the observed episode rate in six probability bands, validation years.

It is calibrated within 0.032 in every probability band (Fig. 5). Clearing remains the hardest case, with an AUC of only 0.734.

TABLE VIII
GATE QUALITY, VALIDATION YEARS

	Gate p	PM2.5 now alone
AUC, all hours	0.923	0.904
AUC, onsets (1 688 episode hours)	0.883	0.846
AUC, origin above 150 (clearings)	0.734	0.645
Brier score	0.088	

C. Results

Table IX compares the combinations. The soft split lowers MAE by 0.42 µg m^{-3} (1.2%) and RMSE by 0.07. Rebuilt with three sets of five seeds, the MAE gain is 0.42, 0.42 and 0.46 while the normal expert itself moves by only 0.05 between sets, so it is real; the RMSE gain is 0.07, 0.01 and 0.20, within the 0.12 the normal expert moves, so it is not. A hard switch at $p = 0.5$ is much worse, because confident wrong calls are expensive under squared error. With the true label as gate, the same experts reach RMSE 39.46, 30% below the blend: the experts are good enough, and the gate is the bottleneck.

TABLE IX
MIXTURE OF EXPERTS AT SIX HOURS, VALIDATION YEARS (RMSE AND MAE IN µg m^{-3})

			RMSE on		
Model	RMSE	MAE	episode	onset	clearing
Persistence	65.77	40.23	89.21	108.17	162.63
Normal (blend)	56.33	36.41	87.08	92.42	111.63
Gate, $\lambda = 0.1$	56.31	36.40	85.27	90.95	114.34
Split, soft	**56.26**	**35.99**	86.57	92.64	113.20
Split, hard ($p > 0.5$)	61.24	38.12	89.83	108.19	140.14
Split, true label	39.46	27.29	64.88	54.93	42.26

IX. Discussion and limitations

The mixture result explains why further feature or model tuning brings little. A calibrated soft mixture estimates the same conditional mean as a single regression on the same inputs, so it can only gain where the gate knows more than the experts, and here it does not: the north-west wind that ends an episode often arrives only after the forecast is made, which inputs up to hour $t - 1$ cannot show. Larger gains need more information, such as a longer input history, upstream stations or weather forecasts.

Validation scores are selection scores and slightly optimistic, and RMSE is a noisy selection metric here: with half of the squared error in a fifth of the hours, small differences between models are not reliable, which is why we confirm gains over several seed sets. The data come from one station without PM10, other pollutants or boundary-layer height. The one-hour weather lag guards against an unresolved clock offset between PM2.5 and weather and may cost some accuracy if the clocks agree. The air quality categories are applied to hourly values. Finally, 2014 is a single test year, and an easier one than the validation years.

X. Conclusion and Phase 2

A leak-free gradient boosting pipeline forecasts PM2.5 six hours ahead with 15.1% lower RMSE than persistence on an unseen year, using PM2.5 history, weather and physics-informed stagnation features. The remaining error sits in the onset and clearing of pollution episodes, and a perfect episode gate would cut RMSE by 30%. Phase 2 will train an LSTM or temporal CNN on fixed input windows under the same folds and test protocol, and will test whether a recurrent gate that reads a longer history can supply the information the tree-based gate lacks.

Reproducibility

The code is public at https://github.com/Vetri-78640/aml_dl_pm2.5: a Python package (`src/aqf`) for data, features, models, metrics and figures, one configuration file per stage, four notebooks (EDA, preprocessing, Phase 1 models, mixture of experts) and 22 automated tests covering the data pipeline, causality, the embargo and model behaviour. The Phase 1 notebook runs in about seven minutes on a laptop CPU and the mixture-of-experts notebook in under three; all results are saved as CSV files under `reports/`.

Team contributions

- **Vetriselvan R**: project setup, dataset and documentation; mixture of experts (gate, notebook and model history).
- **Ishan Goyal**: exploratory data analysis on the original data and the combined data report.
- **Vedant Madne**: preprocessing pipeline and tests, preprocessing walkthrough, physics feature group and feature tests.
- **Pampana Sree Adithya**: preprocessing report and figures, skew check on the development years, calendar and humidity feature builders.
- **Hrishi Dongre**: baselines, ridge and XGBoost models, Phase 1 experiments, results and figures.

References

[1] S. Chen, "Beijing PM2.5," UCI Machine Learning Repository, 2015, doi: 10.24432/C5JS49.

[2] X. Liang, T. Zou, B. Guo, S. Li, H. Zhang, S. Zhang, H. Huang, and S. X. Chen, "Assessing Beijing's PM2.5 pollution: severity, weather impact, APEC and winter heating," *Proc. R. Soc. A*, vol. 471, no. 2182, Art. no. 20150257, 2015.

[3] V. R, H. Dongre, P. S. Adithya, I. Goyal, and V. Madne, "Data understanding and leak-free preprocessing for short-horizon PM2.5 forecasting in Beijing," Team T024 project report, 2026.

[4] S. Chen, "PM2.5 data of five Chinese cities," UCI Machine Learning Repository, 2016, doi: 10.24432/C52K58.

[5] J. Zhong, X. Zhang, Y. Dong, Y. Wang, C. Liu, J. Wang, Y. Zhang, and H. Che, "Feedback effects of boundary-layer meteorological factors on cumulative explosive growth of PM2.5 during winter heavy pollution episodes in Beijing from 2013 to 2016," *Atmos. Chem. Phys.*, vol. 18, pp. 247–258, 2018, doi: 10.5194/acp-18-247-2018.

[6] Q. Huang, X. Cai, J. Wang, Y. Song, and T. Zhu, "Climatological study of the Boundary-layer air Stagnation Index for China and its relationship with air pollution," *Atmos. Chem. Phys.*, vol. 18, pp. 7573–7593, 2018, doi: 10.5194/acp-18-7573-2018.

[7] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.

[8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, doi: 10.1145/2939672.2939785.

[9] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[10] Y. Wang and R. Du, "Same-hour estimation and multi-horizon forecasting of PM2.5 in Beijing: Model comparison and feature-group contributions," arXiv:2607.07279, 2026.

[11] S. Ben Taieb, G. Bontempi, A. F. Atiya, and A. Sorjamaa, "A review and comparison of strategies for multi-step ahead time series forecasting based on the NN5 forecasting competition," *Expert Systems with Applications*, vol. 39, no. 8, pp. 7067–7083, 2012.

[12] L. J. Tashman, "Out-of-sample tests of forecasting accuracy: an analysis and review," *Int. J. Forecasting*, vol. 16, no. 4, pp. 437–450, 2000.

[13] C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," *Information Sciences*, vol. 191, pp. 192–213, 2012.

[14] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Trans. Knowl. Discov. Data*, vol. 6, no. 4, Art. no. 15, 2012.

[15] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ, USA: Wiley, 2018.

[16] J. M. Bates and C. W. J. Granger, "The combination of forecasts," *Operational Research Quarterly*, vol. 20, no. 4, pp. 451–468, 1969.

[17] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991.

[18] N. Duan, "Smearing estimate: A nonparametric retransformation method," *J. Amer. Statist. Assoc.*, vol. 78, no. 383, pp. 605–610, 1983.

[19] O. A. Alduchov and R. E. Eskridge, "Improved Magnus form approximation of saturation vapor pressure," *J. Appl. Meteorol.*, vol. 35, no. 4, pp. 601–609, 1996.

[20] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[21] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[22] Ministry of Environmental Protection of China, *Technical Regulation on Ambient Air Quality Index (on trial)*, HJ 633-2012, 2012.